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Status	Finished
Started	Thursday, 16 October 2025, 12:01 PM
Completed	Thursday, 16 October 2025, 12:27 PM
Duration	26 mins 38 secs
Marks	2.30/3.00
Grade	2.30 out of 3.00 (76.67%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.30 out of 1.00

Consider the following EBNF grammar.

$$N \rightarrow e G \mid p P$$

$$G \rightarrow e S \mid v S$$

$$S \rightarrow e I \mid q P$$

$$P \rightarrow e I \mid k I$$

$$I \rightarrow N \mid p$$

Translate it into an NFA, given as 5-tuple.

Use the nonterminal names as the corresponding state names and F for the final state.

Your answers must be syntactically correct for marking.

Sample 5-tuple for an NFA with three states A,B,C with A as start state, C as final state, and three transitions: from A with a to B, from B with b to C, and from C with ϵ to A:

States = {"A","B","C"}

Alphabet = {"a","b"}

Transitions = { ["A","a","B"], ["B","b","C"], ["C","", "A"] }

Start state = "A"

Final States = {"C"}

States= {"N","G","P","S","I"}

Alphabet
= {"e","p","k","v","q"}

Transitions
= {["N","e","G"],["N","p","P"],["G","e","S"],["G","v","S"],["S","e","I"],["S","q","P"],["P","e","I"],["P","k","I"],["I","", "N"]}

Start state
= "N"

Final
States = {"I"}

Your last answer was interpreted as follows:

{N, G, P, S, I}

Your last answer was interpreted as follows:

{e, p, k, v, q}

Your last answer was interpreted as follows:

{[N, e, G] , [N, p, P] , [G, e, S] , [G, v, S] , [S, e, I] , [S, q, P] , [P, e, I] , [P, k, I] , [I, , N]}

Your last answer was interpreted as follows:

N

Your last answer was interpreted as follows:

{I}

Question **2**

Correct

Mark 1.00 out of 1.00

Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant G of type Nat with the value 5.

Specification statement: $\{c \geq 19\} u, w, c := u', w', c' \mid w' \leq c \Rightarrow u' \leq 7$

Refinement rule Introduce Constant (InC)

$\{\text{pre}\} v, x := v', x' \mid \text{rel}$
 $[=$
 $\text{CON } C:T \mid C=x \cdot \{C=x \text{ and pre}\} v, x := v', x' \mid \text{rel}$

Fill in the correctly refined specification statement. You can copy parts of the original statement.

$\text{CON } G:\text{Nat} \mid G=5 \cdot \{G=5 \text{ and } c \geq 19\} u, w, c := u', w', c' \mid w' \leq c \Rightarrow u' \leq 7$

Your last answer was interpreted as follows:

$\text{CON } G:\text{Nat} \mid G=5 \cdot \{G=5 \text{ and } c \geq 19\} u, w, c := u', w', c' \mid w' \leq c \Rightarrow u' \leq 7$

Question **3**

Correct

Mark 1.00 out of 1.00

Consider the following relation R over integers.

xRy iff $x \cdot x + y \cdot y \geq 4$

Check the properties of the relation R .

The relation is reflexive:

The relation is symmetric:

The grammar is antisymmetric:

The grammar is transitive:

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 359

Signing code:

Your last answer was interpreted as follows:

34670

[◀ Test 1 \(topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars\)](#)

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